

Session 11 — Stretch --- Commodities then Currencies

In class (90 min). Type along with the instructor. The project is **not graded**. The final exam is a closed-book MCQ: you must be able to **explain every line** you type.

Goal. Optional: futures return snapshot, then a G8 FX matrix and one pair chart.

You leave with. Whatever you finish. Skipping this session's stretch is fine for the exam if Stocks and Bonds work.

This session is **stretch**. If last week is incomplete, stay on Bonds (empty states, captions, last-updated). Session 12 will not add tabs.

Order: Commodities first (reuses `download_data` and return tables). Currencies second (new FX logic). Do not start FX if commodities is still messy.

```
uv run streamlit run dashboard.py
```

Commodities

Universe (Yahoo futures):

- Metals: GC=F Gold, SI=F Silver, HG=F Copper
- Energy: CL=F WTI, BZ=F Brent, NG=F Natural gas
- Grains: ZW=F Wheat, ZC=F Corn

Bundled files look like GC_F.csv (= becomes _).

Sidebar keys: `commodities_lookback`, `commodities_start_date`, `commodities_end_date`, plus a multiselect of **names** (default: all).

Fetch extra history like Stocks (end - 365 days) so 1-year returns work.

For each selected ticker that has a column in prices:

- `compute_return_metrics(prices[ticker])`
- `st.subheader` with the human name
- `render_return_metrics(...)`

Then two 2×2 bar charts (trailing horizons, then calendar horizons). Each subplot is one horizon; bars are the selected names. Use `plotly.subplots.make_subplots(rows=2, cols=2, ...)`.

If time is short: **tables only**, skip the bar grids.

```
from datetime import date, timedelta

import plotly.graph_objects as go
import streamlit as st
from plotly.subplots import make_subplots

from views.common import (
    CALENDAR_HORIZONS,
    DEFAULT_END_DATE,
    LOOKBACK_YEARS,
    TRAILING_HORIZONS,
    chart_layout,
    compute_return_metrics,
    date_range_error,
    download_data,
    get_available_date_bounds,
    lookback_start,
    render_return_metrics,
)
```

Add to `common.py` if missing:

```
TRAILING_HORIZONS = ("7-day", "30-day", "90-day", "1-year")
CALENDAR_HORIZONS = (
    "Week to date",
    "Month to date",
    "Quarter to date",
    "Year to date",
)
```

No price-path line charts on this tab. No regime colours.

Currencies

G8: USD, EUR, GBP, JPY, CHF, AUD, CAD, NZD.

Yahoo **USD legs** (not every cross):

```
USD_LEGS = {
    "EUR": ("EURUSD=X", "usd_quote"),
    "GBP": ("GBPUSD=X", "usd_quote"),
    "AUD": ("AUDUSD=X", "usd_quote"),
    "NZD": ("NZDUSD=X", "usd_quote"),
    "JPY": ("USDJPY=X", "usd_base"),
    "CHF": ("USDCHF=X", "usd_base"),
    "CAD": ("USDCAD=X", "usd_base"),
}
```

usd_quote means the ticker is already USD per 1 foreign unit. usd_base means foreign per 1 USD, so invert: $1 / \text{price}$.

```
def usd_per_unit(leg_price: float, convention: str) -> float:
    if convention == "usd_quote":
        return float(leg_price)
    return 1.0 / float(leg_price)
```

Build daily USD-per-unit paths, then a pair:

```
def pair_rate(usd_paths, numerator: str, denominator: str):
    name = f"{numerator}/{denominator}"
    if numerator == denominator:
        return pd.Series(1.0, index=usd_paths.index, name=name)
    return (usd_paths[numerator] / usd_paths[denominator]).dropna().rename(name)
```

Spot matrix (latest day): units of **column** per 1 unit of **row** = usd_per[row] / usd_per[col].

Sidebar keys: currencies_lookback, currencies_start_date, currencies_end_date, plus numerator and denominator selectboxes.

Download FX_TICKERS with download_data. Bundled files: EURUSD_X.csv, etc.

Enough for class: st.dataframe of the matrix + one Plotly line for the selected pair. Clickable matrix cells are optional.

No correlation heatmap. No overlays on this tab.

Commit whatever you have

```
git add views/commodities.py views/currencies.py views/common.py
git commit -m "session 11: stretch tabs"
```

Next week

Feature freeze. We deploy and practise **reading** code. Do not open new features.